

- 2 The responses of plant species to water stress can be classified as either isohydric or anisohydric.
- Isohydric plant species close stomata during times of water stress. This behaviour minimises water loss by transpiration but also reduces carbon dioxide uptake for photosynthesis.
 - Anisohydric plant species do not close stomata during times of water stress. This behaviour maximises carbon dioxide uptake for photosynthesis but also increases water loss by transpiration.

A biologist studied 10 tree species from Australia. The biologist studied the effect of water stress **and** high environmental temperatures on five isohydric tree species and five anisohydric tree species.

For each tree species studied:

- The biologist obtained 20 young trees.
 - The young trees were grown in containers of soil in controlled conditions in a glasshouse.
 - The environmental conditions in the glasshouse were chosen to represent summer conditions in Australia. The mean glasshouse temperature was 28 °C.
 - All the young trees were given a good supply of water for 10 weeks, so that the young trees acclimatised to the environmental conditions in the glasshouse.
- (a) State **three** environmental conditions in the glasshouse that should be standardised in the 10-week period of acclimatisation, **other than** the temperature of the glasshouse.

.....

.....

.....

.....

.....

.....

..... [3]

After the 10-week period of acclimatisation, the biologist divided the young trees from each species into four groups of five trees.

Table 2.1 shows the experimental conditions used by the biologist for the next five weeks.

Table 2.1

group	experimental conditions
1	The young trees were given a good supply of water for five weeks. The mean glasshouse temperature during weeks 1 to 5 was 28 °C.
2	The young trees were given a good supply of water for five weeks. The mean glasshouse temperature during weeks 1 to 4 was 28 °C. During week 5, the mean glasshouse temperature was increased to 35 °C.
3	The young trees were given a reduced supply of water for five weeks (water stress). The mean glasshouse temperature during weeks 1 to 5 was 28 °C.
4	The young trees were given a reduced supply of water for five weeks (water stress). The mean glasshouse temperature during weeks 1 to 4 was 28 °C. During week 5, the mean glasshouse temperature was increased to 35 °C.

At the end of week 5, the biologist measured the stomatal conductance of three leaves from each young tree at 12:00 (midday).

Stomatal conductance is a measure of water vapour loss from the intercellular air spaces of leaves to the atmosphere through the stomata.

The biologist processed the data to compare the results from the isohydric and anisohydric tree species in the four experimental conditions, as shown in Table 2.1.

The results are shown in Fig. 2.1.

Key

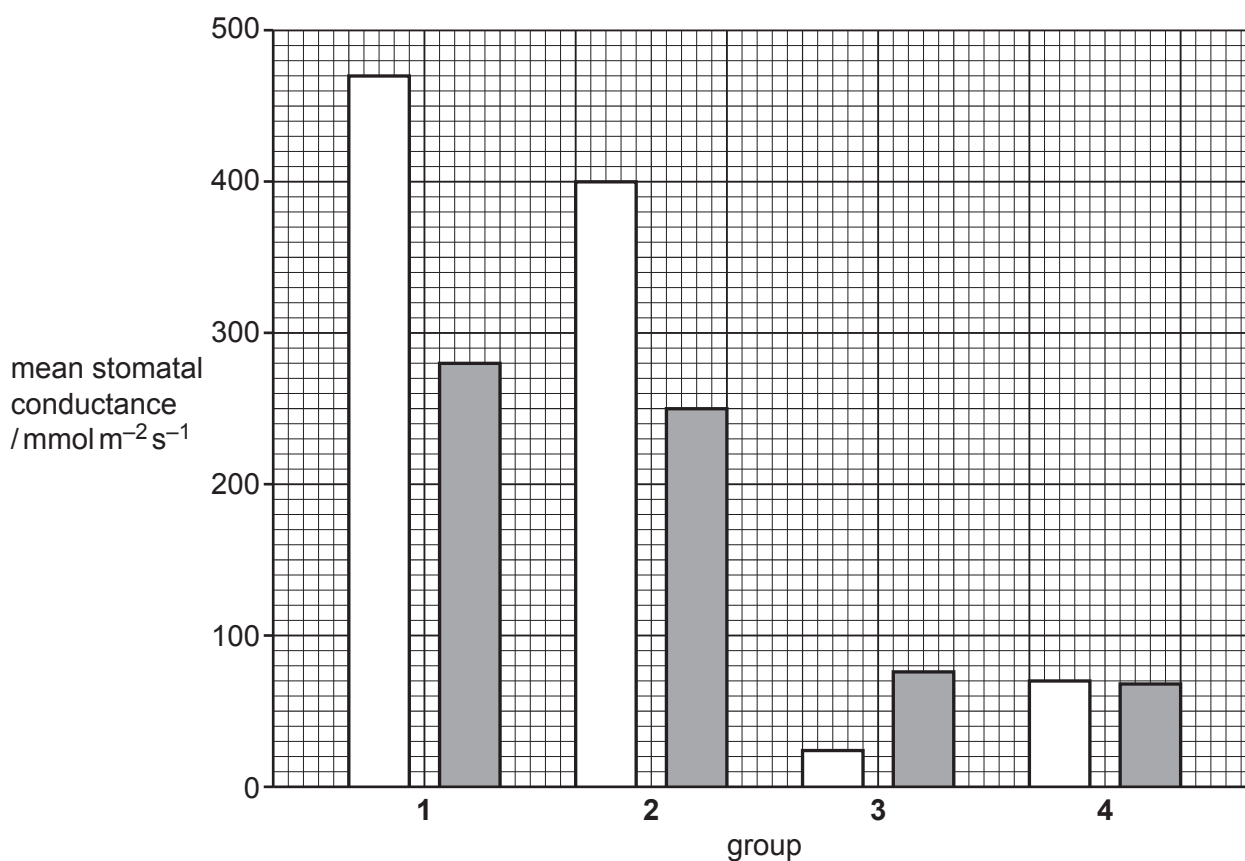
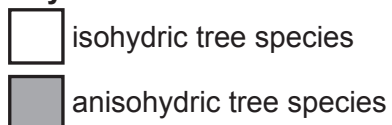


Fig. 2.1

- (b) Using Fig. 2.1, state the effect of a high mean temperature on the mean stomatal conductance of young trees that were given a good supply of water.

.....

.....

..... [1]

- (c) The biologist then compared the young trees from group 3 and group 4 that were exposed to water stress.

The biologist carried out statistical tests on the data to see if the difference between the mean stomatal conductance of young trees in group **3** and the mean stomatal conductance of young trees in group **4** was significant.

Table 2.2 shows the probability values (p) from the results of the statistical tests.

Table 2.2

tree species	value of p	significance
isohydric tree species	0.046	significant
anisohydric tree species	0.788	not significant

With reference to Fig. 2.1 and Table 2.2, suggest **and** explain the conclusions that can be made about the effect of water stress **and** a high environmental temperature on isohydric **and** anisohydric tree species.

..... [4]

Question 2 continues on page 12.

- (d) The biologist noticed that the young trees in group 4 had some dead leaves at the end of week 5.

The biologist determined the percentage of leaves on the young trees that were dead at the end of week 5.

The results from the isohydric tree species and the anisohydric tree species are shown in Table 2.3.

Table 2.3

type of tree species	group 4: water stress + high mean temperature	
	mean percentage of leaves that were dead	standard error (SE)
isohydric	19.0	8.3
anisohydric	3.5	1.3

- (i) The biologist then analysed these data using a t -test to compare the mean percentage of leaves that were dead on the young trees of the isohydric and anisohydric species.

State a null hypothesis for the t -test.

.....

 [1]

- (ii) The formula for the t -test is:

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

key to symbols:

$\bar{x}$ = mean

s = sample standard deviation

n = sample size (number of observations)

The biologist calculated $\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)} = 8.231$

Use this value and Table 2.3 to calculate the value of t for these data.

Show your working.

$t =$ [2]

- (iii) The degrees of freedom for this t -test are 48. Table 2.4 shows the probability table for the t -test.

Table 2.4

degrees of freedom	critical values		
	$p = 0.10$ (10%)	$p = 0.05$ (5%)	$p = 0.01$ (1%)
48	1.677	2.011	2.682

Using Table 2.4 and the calculated value of t from (d)(ii), describe what the biologist can conclude from the results shown in Table 2.3.

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 14]